

# Samuel Tilsen

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## Curriculum Vitae

### Academic appointments

|           |                                                                                                           |
|-----------|-----------------------------------------------------------------------------------------------------------|
| 2023-     | Professor, Cornell University, Department of Linguistics                                                  |
| 2017-     | Associate Professor, Cornell University, Department of Linguistics                                        |
| 2017-2018 | Visiting Scholar, Wolfson College, Oxford University                                                      |
| 2011-2017 | Assistant Professor, Cornell University, Department of Linguistics                                        |
| 2009-2011 | NIH NRSA Postdoctoral Fellow, University of Southern California<br>Hearing and Communication Neuroscience |

### Education

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|------|-----------------------------------------------------------------------------------------------------------------------|
| 2009 | Ph.D. Linguistics, University of California, Berkeley<br>Dissertation: Interactions between speech rhythm and gesture |
| 2006 | M.A. Linguistics, University of California, Berkeley                                                                  |
| 2001 | B.A. Linguistics, B.A. English, Northwestern University                                                               |

### Fellowships, awards, grants

|      |                                                                                                                                                  |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 2022 | (Co-PI) Cornell University New Frontiers Grant: Emergence of dialects in networks of speakers with random, constrained interactions (\$198,558)  |
| 2016 | (Co-PI) Mario Einaudi Center for International Studies Small Grant: LabPhon15: Speech Dynamics and Phonological Representation (\$5,000).        |
| 2016 | (Co-PI) Cornell University Institute for the Social Sciences Small Grant: LabPhon15 - Speech Dynamics and Phonological Representation (\$3,500). |
| 2016 | (Co-PI) NSF Conference Grant: LabPhon 15 - Speech Dynamics and Phonological Representation, Award #1551588, (\$24,845).                          |
| 2013 | Cornell Center for Teaching Excellence Faculty Grant: Earmaster: a phonetic transcription game.                                                  |
| 2012 | Cornell University Institute for the Social Sciences Small Grant: Visualizing speech: real-time MRI of the vocal tract (\$10,680).               |
| 2009 | NIH Ruth L. Kirschstein-NRSA Institutional Research Training Grant, University of Southern California.                                           |
| 2008 | NSF Doctoral Dissertation Improvement Grant, Award #0817767.                                                                                     |
| 2001 | Daniel Bonbright Award, Weinberg College of Arts and Sciences, Northwestern University.                                                          |

### Books

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|------|----------------------------------------------------------------------------------------------|
| 2019 | <b>Syntax with oscillators and energy levels.</b> Tilsen, S. <i>Language Science Press</i> . |
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### Peer-reviewed publications

|      |                                                                                                                                                                                                                                                                                      |
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| 2026 | <b>Phrasal rhythmicity and the sources of temporal intermittency in speech.</b> Tilsen, S. <i>Rhythms of Speech and Language</i> , ch. 20. Cambridge University Press.                                                                                                               |
| 2025 | <b>The Gesture-Field-Register (GFR) framework for modeling F0 control.</b> Kim, S.-E. & Tilsen, S. <i>Journal of Phonetics</i> , doi: 10.1016/j.wocn.2025.101410.                                                                                                                    |
| 2025 | <b>Thai speakers time lexical tones to supralaryngeal articulatory events.</b> Burrioni, F. & Tilsen, S. <i>Journal of Phonetics</i> , doi: 10.1016/j.wocn.2024.101389.                                                                                                              |
| 2024 | <b>Planning for the future and reacting to the present: Proactive and reactive F0 adjustments in speech.</b> Kim, S.-E. & Tilsen, S. <i>Journal of Phonetics</i> , doi: 10.1016/j.wocn.2024.101322.                                                                                  |
| 2024 | <b>Internal speech is faster than external speech: evidence for feedback-based temporal control.</b> Tilsen, S. <i>Cognition</i> , doi: 10.1016/j.cognition.2023.105713.                                                                                                             |
| 2023 | <b>Probability and randomness in phonology: Deep vs. shallow stochasticity.</b> Tilsen, S. <i>Studies in Phonetics, Phonology, and Morphology</i> doi: 10.17959/sppm.2023.29.2.247.                                                                                                  |
| 2023 | <b>Looking within events: examining internal temporal structure with local relative rate.</b> Tilsen, S. & Tiede, M. <i>Journal of Phonetics</i> , doi: 10.1016/j.wocn.2023.101264.                                                                                                  |
| 2023 | <b>Parameters of unit-based measures of speech rate.</b> Tilsen, S. & Tiede, M. <i>Speech Communication</i> doi: 10.1016/j.specom.2023.05.006.                                                                                                                                       |
| 2023 | <b>Hierarchical distinctions in the production and perception of nuclear tones in American English.</b> Cole, J., Steffman, J., Shattuck-Hufnagel, S., & Tilsen, S. <i>Laboratory Phonology</i> doi: 10.16995/labphon.9437.                                                          |
| 2023 | <b>A preliminary study of speech rhythm differences as markers of stuttering persistence in pre-school age children.</b> Erdemir, A., Walden, T., Mefferd, A., Tilsen, S., & Jones, R. M. <i>Journal of Speech, Language, and Hearing Research</i> doi: 10.1044/2022_JSLHR-22-00126. |

- 2022 **An informal logic of feedback-based temporal control.** Tilsen, S. *Frontiers in Human Neuroscience* doi: 10.3389/fnhum.2022.851991.
- 2022 **Relaxation, Percolation, and Non-Spontaneous Fluctuation of Linguistic Behavior in a Quasi-Isolated System.** Tilsen, S. *Frontiers in Physics, Social Physics* doi: 10.3389/fphy.2022.801740.
- 2022 **An investigation of functional relations between speech rate and phonetic variables.** Kim, S.-E. & Tilsen, S. *Journal of Phonetics*, doi: 10.1016/j.wocn.2022.101152.
- 2022 **The online effect of clash is durational lengthening, not prominence shift: Evidence from Italian.** Burroni, F. & Tilsen, S. *Journal of Phonetics*, 91. doi:10.1016/j.wocn.2021.101124.
- 2021 **Localizing category-related information in speech with multi-scale analyses.** Tilsen, S., Kim, S.-E., & Wang, C. *PLoS ONE* 16(1): e0258178.
- 2021 **Temporal differences between high vowels and glides are more robust than spatial differences.** Burgdorf, D. & Tilsen, S. *Journal of Phonetics*, 88. doi:10.1016/j.wocn.2021.101073.
- 2020 **Detecting anticipatory information in speech with signal chopping.** Tilsen, S. *Journal of Phonetics*, 82. doi:10.1016/j.wocn.2020.100996.
- 2020 **Incongruencies between phonological theory and phonetic measurement.** Mucke, D., Hermes, A., & Tilsen, S. *Phonology*, 37: 1, 133-170. doi:10.1017/S0952675720000068.
- 2019 **Motoric mechanisms for the emergence of non-local phonological patterns.** Tilsen, S. *Frontiers in Psychology*. DOI:10.3389/fpsyg.2019.02143.
- 2019 **Space and time in models of speech rhythm.** Tilsen, S. *Annals of The New York Academy of Sciences*. doi:10.1111/nyas.14102.
- 2018 **Analysis of Speech Production Real-time MRI.** Ramanarayanan, V., Tilsen, S., Proctor, M., Toger, J., Goldstein, L., & Nayak, K., Narayanan, S. *Computer Speech & Language*. 52: 1-22. doi:10.1016/j.csl.2018.04.002.
- 2017 **Exertive modulation of speech and articulatory phasing.** Tilsen, S. *Journal of Phonetics*. dx.doi.org/10.1016/j.wocn.2017.03.001.
- 2016 **Shared representations underlie metaphonological judgments and speech motor control.** Tilsen, S. & Cohn, A. *Laboratory Phonology: Journal of the Association for Laboratory Phonology*, 7(1): 14, pp. 1-33, doi:10.5334/labphon.52.
- 2016 **Selection and coordination: the articulatory basis for the emergence of phonological structure.** Tilsen, S. *Journal of Phonetics*, 55: 53-77. doi:10.1016/j.wocn.2015.11.005.
- 2016 **Anticipatory Posturing of the Vocal Tract Reveals Dissociation of Speech Movement Plans from Linguistic Units.** Tilsen, S., Spincemaille, P., Xu, B., Doerschuk, P., Luh, W.-M., Feldman, E., Wang, Y. *PLoS ONE* doi:10.1371/journal.pone.0146813.
- 2015 **Real-time articulatory biofeedback with electromagnetic articulography.** Tilsen, S., Das, D., & McKee, B. *Linguistics Vanguard* doi:10.1515/lingvan-2014-1006.
- 2014 **Selection and coordination in temporally constrained production.** Tilsen, S. *Journal of Phonetics*, 44: 26-46.
- 2013 **A dynamical model of hierarchical selection and coordination in speech planning.** Tilsen, S. *PLoS ONE* 8(4): e62800. doi:10.1371/journal.pone.0062800.
- 2013 **Speech rhythm analysis with decomposition of the amplitude envelope: Characterizing rhythmic patterns within and across languages.** Tilsen, S. & Arvaniti, A. *Journal of the Acoustical Society of America*, 134, 1: 628-639. doi:10.1121/1.4807565.
- 2013 **Inhibitory mechanisms in speech planning maintain and maximize contrast.** Tilsen, S. In A. Yu (Ed.), *Origins of Sound Patterns: Approaches to Phonologization*. 112-127. Oxford: Oxford University Press.
- 2012 **Articulatory gestures are individually selected in production.** Tilsen, S. & Goldstein, L. *Journal of Phonetics*, 40: 764-779.
- 2012 **Utterance preparation and stress clash: planning prosodic alternations.** Tilsen, S. In Fuchs, S., Perrier, P., Weirich, M. & Pape, D. (Eds.) *Speech production and perception: Planning and dynamics*, pp. 119-156. Frankfurt Am Main: Peter Lang Verlag.
- 2011 **Effects of syllable stress on articulatory planning observed in a stop-signal experiment.** Tilsen, S. *Journal of Phonetics*, 39: 642-659.
- 2011 **Metrical regularity facilitates speech planning and production.** Tilsen, S. *Laboratory Phonology*, 2: 1, 185-218.
- 2010 **Automated measurement of vowel formants in the Buckeye Corpus.** Yao, Y., Tilsen, S., Sprouse, R., & Johnson, K. *Gengo Kenkyu* (Journal of the Linguistic Society of Japan), 138: 99-113.
- 2009 **Multi-timescale dynamical interactions between speech rhythm and gesture.** Tilsen, S. *Cognitive Science*, 33, 839-879.
- 2009 **Subphonemic and cross-phonemic priming in vowel shadowing: evidence for the involvement of exemplars in production.** Tilsen, S. *Journal of Phonetics*, 37:3, 276-296.
- 2008 **Low-frequency Fourier analysis of speech rhythm.** Tilsen, S. & Johnson, K. *Journal of the Acoustical Society of America*, 124:2, EL34-39.

## Conference proceedings

- 2025 **Envelope-based metrics reveal rhythmic development tendencies in Brazilian L2 English Learners.** Teixeira, L. A. S., Tilsen, S., Lima, R. M., & Arantes, P. *preprint*
- 2023 **EchoNose: Sensing Mouth, Breathing and Tongue Gestures inside Oral Cavity using a Non-contact Nose Interface.** Sun, R., Zhou, X., Steeper, B., Zhang, R., Yin, S., Li, K., Wu, S., Tilsen, S., Guimbretiere, F., Zhang, C. *Proceedings of the 2023 International Symposium on Wearable Computers, Association for Computing Machinery* DOI: 10.1145/3594738.3611358.
- 2023 **Morpho-phonological effects on the phonetic characteristics of tense consonants in Korean compounds.** Kwon, C. D., Tilsen, S., & Whitman, J. *Proc. of the 20th International Congress of Phonetic Sciences*.
- 2023 **Investigations of F0 control: pitch targets vs pitch register.** Kim, S.-E. & Tilsen, S. *Proc. of the 20th International Congress of Phonetic Sciences*.
- 2022 **Shape matters: Machine classification and listeners' perceptual discrimination of American English intonational tunes.** Cole, J., Tilsen, S., & Steffman, J. *Proc. Speech Prosody 2022* 297-301.
- 2021 **Articulatory characteristics of Icelandic voiced fricative lenition: gradience, categoricity, and speaker/gesture-specific effects.** Stefansdottir, B., Burroni, F., & Tilsen, S. *Proc. Interspeech 2021*.

- 2021 **Compensation for altered feedback in vowels and glides.** Burgdorf, D. & Tilsen, S. *Proc. of the 12th International Seminar on Speech Production*.
- 2021 **Temporal localization of syntactically conditioned prosodic information.** Kim, S.-E. & Tilsen, S. *Proc. of the 12th International Seminar on Speech Production*.
- 2021 **Nonlinear effects of speech rate on articulatory timing in singletons and geminates.** Tilsen, S. & Hermes, A. *Proc. of the 12th International Seminar on Speech Production*.
- 2021 **Cross-linguistic timing contrast in geminates: a rate-independent perspective.** Hermes, A., Tilsen, S., & Ridouane, R. *Proc. of the 12th International Seminar on Speech Production*.
- 2020 **Prominence clash does not induce rhythmic adjustments in Italian.** Burroni, F. & Tilsen, S. *Proc. Speech Prosody 2020* 571-575, DOI: 10.21437/SpeechProsody.2020-117.
- 2020 **Speech rate and syntactically conditioned influences on prosodic boundaries.** Kim, S.-E. & Tilsen, S. *Proc. Speech Prosody 2020* 434-438, DOI: 10.21437/SpeechProsody.2020-89.
- 2019 **Strength and structure: coupling tones with oral constriction gestures.** Mucke, D., Hermes, A., & Tilsen, S. *Proc. Interspeech 2019*, 914-918.
- 2016 **A shared control parameter for F0 and intensity.** Tilsen, S. *Proc. Speech Prosody 2016*, 1066-1070.
- 2016 **Early Prosodic Manifestations of Disfluency.** Li, J. & Tilsen, S. *Proc. Speech Prosody 2016*, 1235-1239.
- 2015 **Structured nonstationarity in articulatory timing.** Tilsen, S. *Proc. of the 18th International Congress of Phonetic Sciences*.
- 2015 **Relation between syllable count judgments and durations of English liquid rimes.** Cohn, A. & Tilsen, S. *Proc. of the 18th International Congress of Phonetic Sciences*.
- 2015 **Phonetic evidence for two types of disfluency.** Li, J. & Tilsen, S. *Proc. of the 18th International Congress of Phonetic Sciences*.
- 2015 **The articulatory tone-bearing unit: Gestural coordination of lexical tone in Thai.** Karlin, R. & Tilsen, S. *POMA*, 22, 060006.
- 2015 **A gestural account of mandarin tone sandhi.** Yi, H. & Tilsen, S. *POMA*, 22, 060003.
- 2015 **A real-time MRI investigation of anticipatory posturing in prepared responses.** Tilsen, S., Spincemaille, P., Xu, B., Doerschuk, P., Karlin, R., Yi, H., Feldman, E., Luh, W., Wang, Y. *POMA*, 22, 060001.
- 2014 **Some people are eager to be heard: anticipatory posturing in speech production.** Tilsen, S., Spincemaille, P., Xu, B., Doerschuk, P., Karlin, R., Yi, H., Feldman, E., Luh, W., Wang, Y. *168th Meeting of the Acoustical Society of America Lay Language Papers*.
- 2013 **Comparison of native and non-native consonant articulation with real-time magnetic resonance imaging of the vocal tract.** Tilsen, S., Xu, B., Spincemaille, P., Srivastava, M., Doerschuk, P., Wang, Y. *POMA*, 19, 060246.
- 2013 **Acoustic and articulatory evidence for the phonological status of liaison consonants.** L'Esperance, M.-J. & Tilsen, S. *POMA*, 19, 060245.
- 2008 **Relations between speech rhythm and segmental deletion.** Tilsen, S. *Proc. of the Annual Meeting of the Chicago Linguistic Society*, 44:1, 221-223.

## Working papers

- 2022 **Interpreting phonetic evidence for hierarchical organization of prosodic phrases.** Kim, S.-E. & Tilsen, S. *Cornell Working Papers in Phonetics and Phonology*, 2022.
- 2020 **Exploratory investigations of SVO sentence production: evidence for category-specific interference effects.** Tilsen, S. *Cornell Working Papers in Phonetics and Phonology* 2020.
- 2020 **A different view of gestural activation: learning gestural parameters and activation with an RNN.** Tilsen, S. *Cornell Working Papers in Phonetics and Phonology* 2020.
- 2018 **Three mechanisms for modeling articulation: selection, coordination, and intention.** Tilsen, S. *Cornell Working Papers in Phonetics and Phonology* 2018.
- 2016 **Correlations between speech behavior and social network structure in a constrained vocabulary game.** Tilsen, S. *Cornell Working Papers in Phonetics and Phonology* 2016.
- 2015 **Speech and social network dynamics in a constrained vocabulary game: design and hypotheses.** Tilsen, S. *Cornell Working Papers in Phonetics and Phonology* 2015.
- 2014 **Syllable count judgments and durations of liquid rimes in English.** Tilsen, S., Cohn, A., & Ricciardi, E. *Cornell Working Papers in Phonetics and Phonology* 2014.
- 2014 **Selection-coordination theory.** Tilsen, S. *Cornell Working Papers in Phonetics and Phonology* 2014.
- 2013 **Imitation of intonational gestures: a preliminary report.** Tilsen, S., Burgess, D., & Lantz, E. *Cornell Working Papers in Phonetics and Phonology* 2013.
- 2012 **Acoustic analysis of the effects of metrical regularity on interval durations.** Tilsen, S., Cibelli, E., Kang, S., Houde, J., Nagarajan, S. *Cornell Working Papers in Phonetics and Phonology* 2012.
- 2012 **A cross-linguistic investigation of articulatory coordination in word-initial consonant clusters.** Tilsen, S., Zec, D., Bjorndahl, C., Butler, B., L'Esperance, M. J., Fisher, A., Heimisdottir, L., Renwick, M., Sanker, C. *Cornell Working Papers in Phonetics and Phonology* 2012.
- 2009 **Toward a dynamical interpretation of hierarchical linguistic structure.** Tilsen, S. *UC Berkeley Phonology Lab Annual Report*, 462-512.
- 2008 **Preliminary results of a stop-signal experiment.** Tilsen, S. *UC Berkeley Phonology Lab Annual Report*, 686-712.
- 2007 **Vowel-to-vowel coarticulation and dissimilation in response-priming.** Tilsen, S. *UC Berkeley Phonology Lab Annual Report*, 416-458.
- 2007 **Low-frequency Fourier analysis of speech rhythm in the VIC corpus.** Tilsen, S. *UC-Berkeley Phonology Lab Annual Report*, 686-712.
- 2006 **Rhythmic coordination in repetition disfluency: a harmonic timing effect.** Tilsen, S. *UC Berkeley Phonology Lab Annual Report*, 73-114.